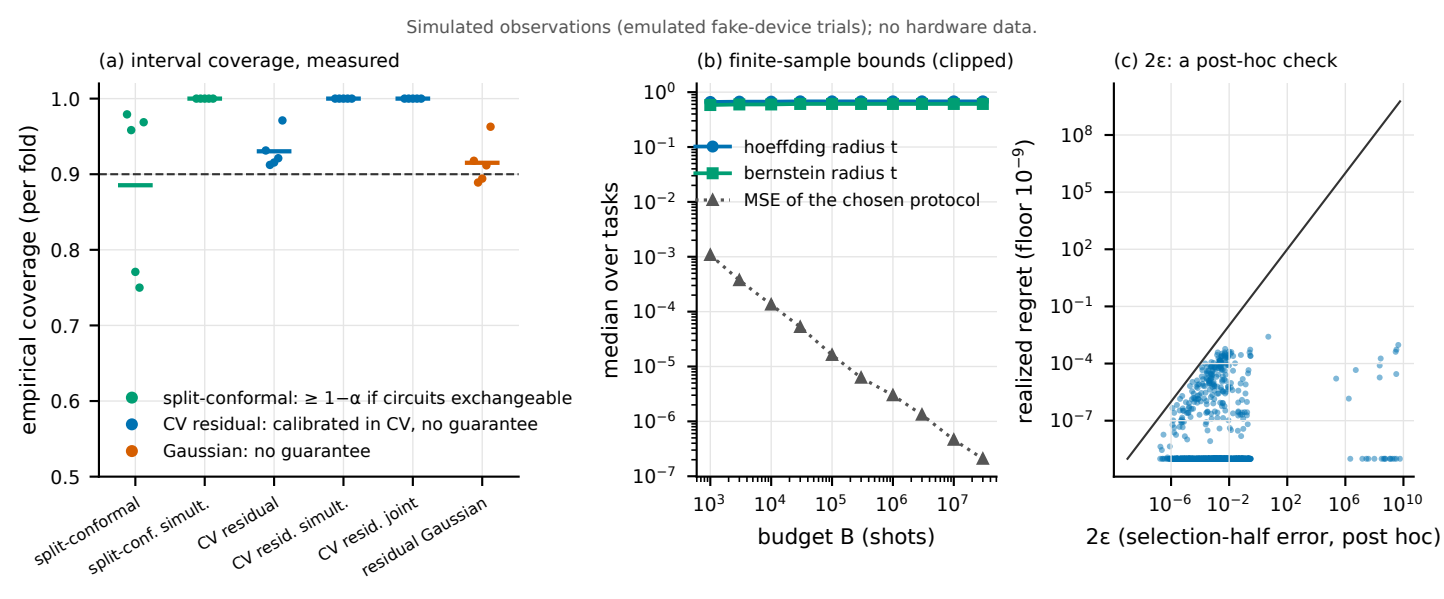
Simulated observations (emulated fake-device trials); no hardware data.
(a) interval coverage, measured
empirical coverage (per fold)
1.0
0.9
0.8
0.7
0.6
0.5
split-conformal: ≥ 1−α if circuits exchangeable
CV residual: calibrated in CV, no guarantee
Gaussian: no guarantee
split-conformal
split-conf. simult.
CV residual
CV resid. simult.
CV resid. joint
residual Gaussian
(b) finite-sample bounds (clipped)
median over tasks
hoeffding radius t
bernstein radius t
MSE of the chosen protocol
budget B (shots)
(c) 2ε: a post-hoc check
realized regret (floor 10⁻⁹)
2ε (selection-half error, post hoc)